

**BOSCH**

Development Changes in PD-Phase

Edition 05

Confidential

产品开发更改

Doc. Approval: RBCD/EIS

Effective date: 20250610

DC No. / 开发更改编号: PDECR_0016

Date / 日期: 6/19/2025

Step 1: Change request & Basic information

Customer project Name/ 客户项目名称:	57cc e-compressor	MCR No. / MCR号:	BM-00095101_003
Affect Product No. / 产品号:	Component No. / 零部件号:		
所有57cc e-compressor成品		F01Z7A008T-01& 见如下:	
Sample type / 样品类型:	☐ A ☐ B ☒ C ☒ FD	Initiator/发起人:	Fan Change (RBCD/EIS4)

Reason of changes / 更改理由:

1.将POE(RL68H)冰熊冷冻油加入BOM,纳入直采范围 (详见: 2025/5/6 e-Compressor SEC core meeting)

2.统一相关文件中的冷冻油的名称和相关铭牌定义形式

3.BOM 中data sheet 位置号删除, 统一装配图纸和优化BOM 形式。

4: 更新Qingling F01Z7P0014-02 的注油量

Customer request / 客户要求 ☐	Supplier request / 供应商请求 ☐	Design optimization / 设计优化 ☐
Correction / 更正(错误) ☐	Process / 工艺 ☐	Other / 其它 ☒
		BOM优化 ☒

Step 2: Change proposal & Change discription

Before Change

1: BOM

无冷冻油物料直采要求。

After Change

更新后: Bom 中增加冷冻油物料, 及其对应的加注量。
冷冻油加注量将与BOM一致最终体现在铭牌上。

396.0	230.000ML	F01Z.7A0.08T-01	Oil: Refrigerant-EMDAPTE POE(RL68H)
300.1	1,000 PC	F01Z.7A0.04N-01	Wire/ Grounding wire
300.2	4,000 PC	F01Z.7A0.05E-01	Damper
927.0			IMSS ID-TBD
940.0		F01Z7A0026-01	Nameplate
940.1			Oil: POE (RL68H) 230ml
940.2			Refrigerant: R134a

如下内容为文档更新

2: 标准版Offer drawing 上定义冷冻油型号和加注量参见TCD
(Geely 提出特殊需求, 将冷冻油型号和加注量定义在图纸上)

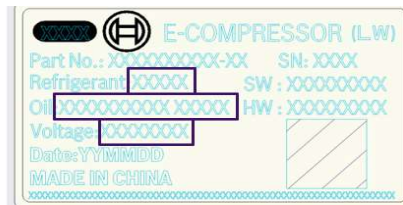
Ambient templateure and use temperature refer to TCD 压缩机环境温度和使用温度范围参见TCD
Type of refrigerant refer to TCD 制冷剂类型参见TCD
Type of refrigerant oil and filling quantity refer to TCD 冷冻油型号和加注量参见TCD

Geely offer drawing 统一成POE(RL68H)

Geely

Type of refrigerant oil and filling quantity:POE68H 230ml
冷冻油型号和加注量: POE68H 230ml

3:铭牌定义如下:

铭牌及其对应的offer drawing 统一成POE(RL68H)
仅示意 (所有涉及料号均需更新)

- 由于不同客户之间注油量不同, 具体项目参考Bom: 940.1

940.1

Refrigerant Oil: 230ml

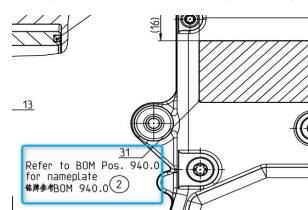
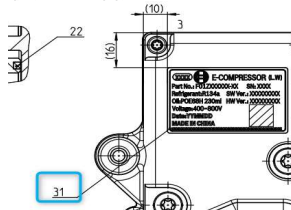
- 为避免铭牌变更带来的客户图纸不断变化。冷冻油/冷媒/电压均参考BOM

- 客户图纸签署时, 提供单独的铭牌示意给客户签字

940.1
940.2
940.3Oil: POE (RL68H) 230ml
Refrigerant: R134a
Voltage: 400V~800V

4: 装配图上使用各个客户的铭牌。

客户无特殊要求的产品, 将统一使用一致的装配图



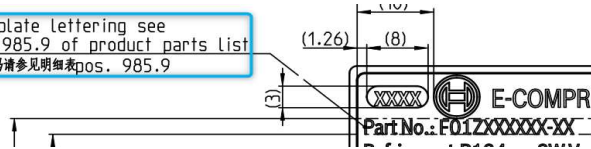
变更理由:为降低维护effort,除客户特殊需求之外,不同客户项目之间可使用统一的装配图。

Initiator / 发起人:	Date / 日期:
Signature / 签名:	Date / 日期:
Check / 校对工程师	Date / 日期:
Signature / 签名:	Date / 日期:
EPQ	Date / 日期:
Signature / 签名:	Date / 日期:

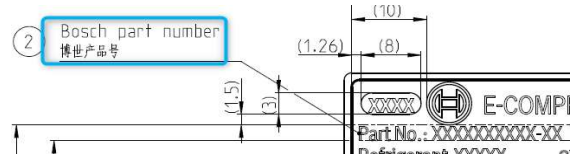
Step 2: Change proposal & Change discription

Before Change

Nameplate lettering see pos. 985.9 of product parts list
铭牌号码请参见明细表pos. 985.9



After Change



5: F01Z7S0003-01 功能检测规范

3.1.3 Test procedure/测试步骤

1. E-compressor code scanning/压缩机扫码
2. Weigh E-compressor/称重: W₀. Fill E-compressor with Refrigerant Oil Emkarate RL68H 230mL. Weigh E-compressor: W₁. 注入 230mL RL68H 冷冻油, 称重 W₁.
3. Assembly E-compressor to function test bench. 安装压缩机至功能台架.
4. Connect E-compressor inlet with air supply machine. 连接压缩机进气口与供气设备.
5. Connect E-compressor electrical connector. 连接插头.
6. Power on and running E-compressor. 上电启动压缩机.
7. Monitor and record related parameters. 监控和记录相关参数.
8. Stop test. Weigh E-compressor after test W₂. 停机, 试验后称重 W₂.

统一成POE(RL68H) 并添加补油描述和注油量公差

3.1.3 Test procedure/测试步骤

1. E-compressor code scanning/压缩机扫码
2. Weigh E-compressor/称重: W₀. Fill E-compressor with Refrigerant Oil POE(RL68H) 230mL(±5%, density 0.98g/mL). Weigh E-compressor: W₁. 注入 POE(RL68H)230mL 冷冻油(公差±5%, 密度 0.98g/mL), 称重 W₁.
3. Assembly E-compressor to function test bench. 安装压缩机至功能台架.
4. Connect E-compressor inlet with air supply machine. 连接压缩机进气口与供气设备.
5. Connect E-compressor electrical connector. 连接插头.
6. Power on and running E-compressor. 上电启动压缩机.
7. Monitor and record related parameters. 监控和记录相关参数.
8. Stop test. Oil replenishment to delivery status as BOM940.1 requirement. 停机, 压缩机补油至发货状态注油量要求 (参照 BOM940.1, 公差±5%, 密度 0.98g/mL).

6: TCD中定义冷冻油类型:

7: 目前BOM中992.0 位置预留了datasheet 的位置号

Bom 中TCD 位置号997.9

997.0	F01Z7D0012-01	OFFER DRAWING
997.9	F01Z7S0001-01	TCD

TCD 中涉及冷冻油区域更新成POE(RL68H)

删除此992.0data sheet区域 (经team 讨论, 无需创建单独的datasheet, 故删除)
更新后TCD 号以Text 形式存在

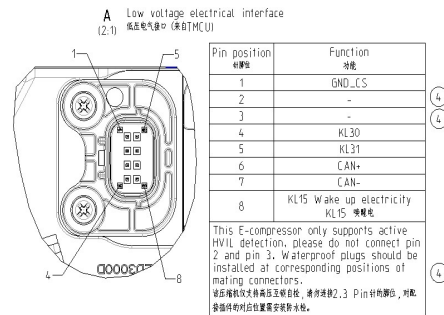
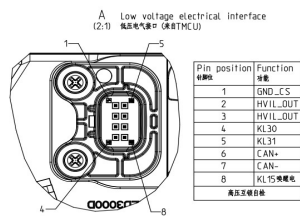
997.0	F01Z7D000H-01	OFFER DRAWING
	TCD: F01Z7S0010-01	

8: offer drawing 增加追溯码备注和描述:

All DMC and traceability codes 所有二维码和追溯码

Except for the nameplate, all DMC and traceability codes are internally used by Bosch, the illustration is only for reference, pls refer to the actual product for accuracy and should not be subject to dimension or visual inspection criteria. 除铭牌外, 所有二维码和追溯码均为博世内部使用, 图示仅供示意, 请以产品为准, 不作为检查项目.

Offer drawing 电气接头信息变更



9: 更新Qingling F01Z7P0014-02 注油量

940.1 Refrigerant Oil: 230ml

940.1 Refrigerant Oil: 250mL

10: 高压互锁检测在ME进行,因此删除总成BOM 下的高压互锁文档

990.2	F01Z7S0008-01	High voltage interlock
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PS: ME的EOL PAV文件 (ME要求的是PAV文档, 非PMB) 已经包含此项检测

Signature / 署名: _____



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Step 5 Documents release / 文档发布

1. Documents release (drawing, offer drawing, Bom, Spec.,) / 文档发布 (图纸, 客户图纸, ...)

Development confirmation/ 研发确认: _____

Date / 日期: _____

Step 6.1 Implementation check list / 导入清单

Departments	Y/N	Description	Responsible	Due date
Development	Y	Change BOMs & Drawings & Documents in POE system	Li Pingzheng	7/15/2025
	Y	Inform documents update (check work-on can met requirements)	Fan Change	7/15/2025
	Y	Update Offer drawing, TCD, D-FMEA	Wang Xiaolong	7/15/2025
	Y	Norm, WB, HF... PMB	Qian Jiong/Sun xudong	7/15/2025
	N	MoC, IMDS.	Li bing/ Li ke	7/15/2025
Manufacturing	N	Related (Production/Testing) equipement be ready on site	N/A	N/A
	N	Related (Production/Testing) program be ready	N/A	N/A
	N	Related (Production/Testing) tooling / cutting / fixture etc. be ready	N/A	N/A
	N	Old tooling / cutting / fixture disposal	N/A	N/A
	N	Old materials disposal	N/A	N/A
	N	Planner update the planning sheet	N/A	N/A
	N	Update FMEA	N/A	N/A
	N	Update CP/FC (Control Plan/Flow Chart)	N/A	N/A
	Y	Update WI/PDS (Include attachments.)	Li Ke	6/30/2025
	N	First batch Mark, Special Mark (Inside Package)	N/A	N/A
	N	First batch Mark, Special Mark (Outside Package)	N/A	N/A
	Y	Training	Li Ke	6/30/2025
COS	Y	Confirm the storage of old parts and coordinate the introduction date for new part (RM)	Meng Fanlong	
	Y	Confirm the delivery date of old parts and first delivery of new parts (FG)		
	N	Check sample orders which affected:material order of CKD	N/A	N/A
	N	Confirm production scheduling according to the alignment, any changes share the information to MOEx,MFE	N/A	N/A
	N	Confirm the old stock/ do prioritize delivery and inventory handling	N/A	N/A
	N	Inform the first delivery to PMO	N/A	N/A
PPS (PPS->PMQ/PQA)	Y	Check sample orders which affected: material order of purchasing	Zheng Xuan	
	Y	Inform internal related departments(COS,MFE,MOEx)with following	Zheng Xuan	
	Y	Update incoming inspection plan	Zheng Xuan	
QMM1	N	Update testing programe on testing equipment	N/A	N/A
	N	Update inspection plan for CKD parts	N/A	N/A
EIS4	Y	Distribute the Offer drawing, TCD to customer	All designer	N/A
LOP	Y	Check 10 digit material order	Ding Yating	
PMO	N	Check sample orders which affected: Customer order	N/A	N/A
	N	Inform Customer the first delivery information	N/A	N/A

Step 6.2 Implementation date / 执行日期

Planned implementation date:

变更计划执行日期: _____

Step 6.3 Implementation approval / 导入清单

Development// 研发	Production/生产	COS/样品	Purchasing/采购	Quality/质量	LOP/ 物流
Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:

Usage list			
		Part No.	Status
1:	冷冻油加注	F01Z7A008T-01	Ongoing
2:	Geely offer drawing 更新	F01Z7D0012-01	7/15/2025
		F01Z7P0026-01	7/15/2025
3:	铭牌更新	F01Z7P008T-01	7/15/2025
		F01Z7P0065-01	7/15/2025
	Bom 更新	All e-compressor	7/15/2025
4	装配图纸更新	F01Z7P0012-01	7/15/2025
		F01Z7P0014-02	7/15/2025
5	功能规范更新	F01Z7S0003-01	7/15/2025
6	TCD 更新	All TCD	7/15/2025
7	Bom 更新	All e-compressor	7/15/2025
9	Offer drawing 更新	F01Z7D0012-01	7/15/2025
		F01Z7D0014-02	7/15/2025
9	注油量更新 BOM	F01Z7P0014-02	Done